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Algorithms and Probability **Theory Sheet T1** **FS 2026**

The solutions for this sheet must be submitted on Moodle by 05 March, 15:59.

THE FOLLOWING EXERCISE IS WORTH 2 BONUS POINTS.

Exercise T1.1 – *Disjoint Paths*

Let $G = (V, E)$ be an undirected graph. For two vertices $s, t \in V$ we denote by $\lambda_{s,t}$ the maximal number of edge-disjoint paths between s and t in G.

(a) Show that for three pairwise different vertices $a, b, c \in V$ we have $\lambda_{a,c} \geq \min\{\lambda_{a,b}, \lambda_{b,c}\}$.

(b) Let $x, y, z \in V$ be three pairwise different vertices and $\alpha, \beta \in \mathbb{N}$ such that $\alpha \leq \lambda_{x,y}$, $\beta \leq \lambda_{x,z}$, and $\alpha + \beta \leq \max\{\lambda_{x,y}, \lambda_{x,z}\}$. Show that there are α many x-y paths and β many x-z-path such that all $\alpha + \beta$ paths are pairwise edge-disjoint.
(Hint: try to construct a graph that contains a vertex v such that the task reduces to finding $\alpha + \beta$ edge-disjoint x-v paths.)

THE FOLLOWING EXERCISE DOES NOT GIVE BONUS POINTS.

Exercise T1.2 – *Connectivity*

(a) Prove or give a counterexample to the following claim: If G is a connected graph such that every vertex of G has even degree, then G is 2-edge-connected.

(b) Prove or give a counterexample to the following claim: If G is a connected graph such that every vertex has degree at least 4, then G is 2-edge-connected.

(c) Prove or give a counterexample to the following claim: For a graph $G = (V, E)$, we define a relation on *vertices* of G: vertices $a \in V$ and $b \in V$ satisfy $a \sim b$ if there is a cycle in G containing both a and b. Then for every graph G relation $\sim$ is an equivalence relation.